ACTL3182 Wk7

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Revision

Definition (Martingale)

Let $M(\cdot)$ be a stochastic process and $\mathcal{F}(t)$ its history up to time t, then $M(\cdot)$ is said to be a $(\mathbb{Q} \,/\, \mathcal{F}_t)$-martingale if

$$\mathbb{E}_{\mathbb{Q}}[M(u) \mid \mathcal{F}(t)] = M(t), \quad \forall u \geq t.$$

That is, given all the information up to time t, the EV of wherever M will be in the future is just what it is right now.

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- A martingale needs two ingredients; the filtration $\mathcal{F}_t$ (the history of a process) and the measure $\mathbb{Q}$.
- Martingales are pure noise processes - their increments are 0 mean.

Martingale Representation Theorem

Previsible Process (discrete)

A stochastic process $\phi(t)$, $t \in \mathbb{N}$, is said to be $\mathcal{F}(t)$-previsible if its value is known from $\mathcal{F}(t-1)$, that is,

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e.g. a trading strategy where at each $t \in \mathbb{N}$, we have the history of stock movements up until time $t-1$ (being $\mathcal{F}(t-1)$) and we have to decide how much to hold at time t, being $\phi(t)$.

Martingale Representation Theorem

Theorem

Let $Y(\cdot)$ and $Z(\cdot)$ be martingales wrt $\mathcal{F}(t)$ and $\mathbb{Q}$. Then, there exists a previsible process $\phi(t)$ such that

$$Y(t) = Y(0) + \sum_{k=1}^{t} \phi(k)(Z(k) - Z(k-1)), \qquad \forall t \in \mathbb{N}.$$

Only use of this is to prove binomial pricing tree and Black Scholes formulas.

Radon-Nikodym derivative (discrete)

Say we have two measures $\mathbb{P}$ and $\mathbb{Q}$ for a discrete random varialbe X and denote

$$\mathbb{P}(X = x_i) = \pi_i, \qquad \mathbb{Q}(X = x_i) = \pi'_i, \quad i = 1, 2, 3, \ldots.$$

The Radon-Nikodym derivative is the random variable Z which takes values

$$Z = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\,\mathbb{P}} = \frac{\mathbb{Q}(X = x_i)}{\mathbb{P}(X = x_i)}, \quad i = 1, 2, 3, \ldots.$$

Brownian Motion

Brownian Motion

A stochastic process $\{W_t : t \geq 0\}$ is said to be a $\mathbb{P}$-Brownian motion if

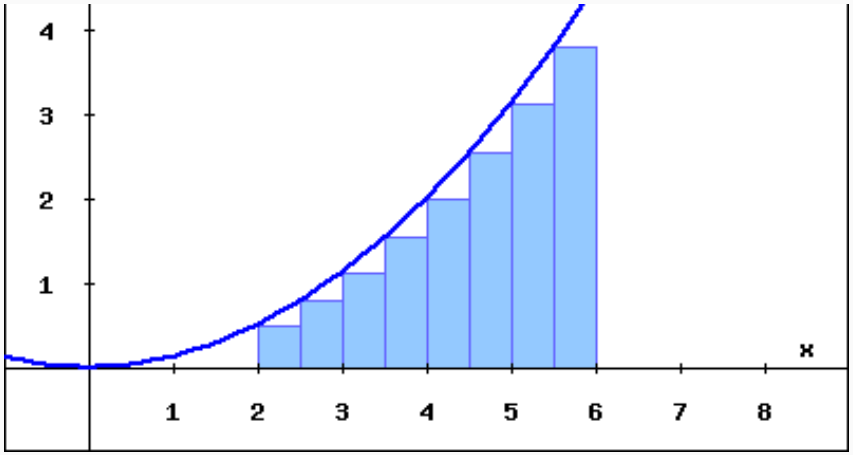
1. W_t has continuous sample paths and $W_0 = 0$;
2. Under $\mathbb{P}$, $W_t \sim \mathcal{N}(0, t)$, $t \geq 0$;
3. Under $\mathbb{P}$, $W_{s+t} - W_s \sim \mathcal{N}(0, t)$ and is independent of $\mathcal{F}_s$, where $\mathcal{F}_s$ is the history of W_t up until time s, $s, t \geq 0$.

What is a (Riemann) integral??

Let f be a function and

$$R_f(t) = \int_0^t f(s)\,\mathrm{d}s = \lim_{\Delta t_i \to 0} \sum_{i=1}^{n} f(t_i^*)\Delta t_i$$

be the Riemann integral process of f.



As we go from t to $t + \delta$, we are really adding $f(t) \cdot \delta$, where δ is a constant taken to be smaller and smaller.

Itô Integral

The Itô integral of f is the exact same idea, but rather than multiplying by the constant $\delta = t_{i+1} - t_i$, we multiply by some random shock $W_{t_{i+1}} - W_{t_i}$,

$$I_f(t) = \int_0^t f(s)\,\mathrm{d}W(s) = \lim_{\delta \to 0} \sum_{i=1}^{n} f(t_i)(W(t_i + \delta) - W(t_i)).$$

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While $R_f(t)$ can be viewed as the accumulation of deterministic drift, $I_f(t)$ can be viewed as the accumulation of random noise.

Itô Process

This motivates the Itô process - for deterministic functions $\mu(\cdot)$ and $\sigma(\cdot)$,

$$X_t = X_0 + \int_0^t \mu(s)\,\mathrm{d}s + \int_0^t \sigma(s)\,\mathrm{d}W_s$$

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Recall that as we go from t to $t + \Delta$, each integral contributes $\mu(t)$ or $\sigma(t)$ multiplied by a shock Δt or ΔW_t. This motivates the shorthand notation,

$$\mathrm{d}X_t = \mu(t)\mathrm{d}t + \sigma(t)\,\mathrm{d}W_t.$$

Itô Lemma

Itô lemma

Let $f(t, x)$ be a function and X_t an Itô process. Then,

$$\mathrm{d}f(t, X_t) = f_t(t, X_t)\,\mathrm{d}t + f_x(t, X_t)\,\mathrm{d}X_t + \frac{1}{2} f_{xx}(t, X_t)\,\mathrm{d}X_t^2$$

Combined with the heuristics

$$(\mathrm{d}t)^2 = 0, \quad (\mathrm{d}t)(\mathrm{d}W_t) = 0, \quad (\mathrm{d}W_t)^2 = \mathrm{d}t,$$

this can be used to find dynamics of functions of Itô processes.